Chapter 4

Linear Models

4.1 Introduction

The determination of the MVU estimator is *in general* a difficult task. It is fortunate, however, that a large number of signal processing estimation problems can be represented by a data model that allows us to easily determine this estimator. This class of models is the *linear model*. Not only is the MVU estimator immediately evident once the linear model has been identified, but in addition, the statistical performance follows naturally. The key, then, to finding the optimal estimator is in structuring the problem in the linear model form to take advantage of its unique properties.

4.2 Summary

The linear model is defined by (4.8). When this data model can be assumed, the MVU (and also efficient) estimator is given by (4.9), and the covariance matrix by (4.10). A more general model, termed the general linear model, allows the noise to have an arbitrary covariance matrix, as opposed to $\sigma^2\mathbf{I}$ for the linear model. The MVU (and also efficient) estimator for this model is given by (4.25), and its corresponding covariance matrix by (4.26). A final extension allows for known signal components in the data to yield the MVU (and also efficient) estimator of (4.31). The covariance matrix is the same as for the general linear model.

4.3 Definition and Properties

The linear model has already been encountered in the line fitting problem discussed in Example 3.7. Recall that the problem was to fit a straight line through noise corrupted data. As our model of the data we chose

$$x[n] = A + Bn + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $w[n]$ is WGN and the slope B and intercept A were to be estimated. In matrix notation the model is written more compactly as

$$\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w} \tag{4.1}$$

where

$$\begin{aligned} \mathbf{x} &= [x[0]\, x[1] \ldots x[N-1]]^T \\ \mathbf{w} &= [w[0]\, w[1] \ldots w[N-1]]^T \\ \boldsymbol{\theta} &= [A\, B]^T \end{aligned}$$

and

$$\mathbf{H} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ \vdots & \vdots \\ 1 & N-1 \end{bmatrix}.$$

The matrix $\mathbf{H}$ is a *known* matrix of dimension $N \times 2$ and is referred to as the *observation matrix*. The data $\mathbf{x}$ are observed after $\boldsymbol{\theta}$ is operated upon by $\mathbf{H}$. Note also that the noise vector has the statistical characterization $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$. The data model in (4.1) is termed the *linear model*. In defining the linear model we assume that the noise vector is Gaussian, although other authors use the term more generally for any noise PDF [Graybill 1976].

As discussed in Chapter 3, it is sometimes possible to determine the MVU estimator if the equality constraints of the CRLB theorem are satisfied. From Theorem 3.2, $\hat{\boldsymbol{\theta}} = \mathbf{g}(\mathbf{x})$ will be the MVU estimator if

$$\frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \mathbf{I}(\boldsymbol{\theta})(\mathbf{g}(\mathbf{x}) - \boldsymbol{\theta}) \tag{4.2}$$

for some function $\mathbf{g}$. Furthermore, the covariance matrix of $\hat{\boldsymbol{\theta}}$ will be $\mathbf{I}^{-1}(\boldsymbol{\theta})$. To determine if this condition is satisfied for the linear model of (4.1), we have

$$\begin{aligned} \frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} &= \frac{\partial}{\partial \boldsymbol{\theta}} \left[-\ln(2\pi\sigma^2)^{\frac{N}{2}} - \frac{1}{2\sigma^2}(\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T(\mathbf{x} - \mathbf{H}\boldsymbol{\theta}) \right] \\ &= -\frac{1}{2\sigma^2}\frac{\partial}{\partial \boldsymbol{\theta}} \left[\mathbf{x}^T\mathbf{x} - 2\mathbf{x}^T\mathbf{H}\boldsymbol{\theta} + \boldsymbol{\theta}^T\mathbf{H}^T\mathbf{H}\boldsymbol{\theta} \right]. \end{aligned}$$

Using the identities

$$\begin{aligned} \frac{\partial \mathbf{b}^T\boldsymbol{\theta}}{\partial \boldsymbol{\theta}} &= \mathbf{b} \\ \frac{\partial \boldsymbol{\theta}^T\mathbf{A}\boldsymbol{\theta}}{\partial \boldsymbol{\theta}} &= 2\mathbf{A}\boldsymbol{\theta} \end{aligned} \tag{4.3}$$

for $\mathbf{A}$ a symmetric matrix, we have

$$\frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \frac{1}{\sigma^2}[\mathbf{H}^T\mathbf{x} - \mathbf{H}^T\mathbf{H}\boldsymbol{\theta}].$$

Assuming that $\mathbf{H}^T\mathbf{H}$ is invertible

$$\frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \frac{\mathbf{H}^T\mathbf{H}}{\sigma^2}[(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x} - \boldsymbol{\theta}] \tag{4.4}$$

which is exactly in the form of (4.2) with

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x} \tag{4.5}$$

$$\mathbf{I}(\boldsymbol{\theta}) = \frac{\mathbf{H}^T\mathbf{H}}{\sigma^2}. \tag{4.6}$$

Hence, the MVU estimator of $\boldsymbol{\theta}$ is given by (4.5), and its covariance matrix is

$$\mathbf{C}_{\hat{\theta}} = \mathbf{I}^{-1}(\boldsymbol{\theta}) = \sigma^2(\mathbf{H}^T\mathbf{H})^{-1}. \tag{4.7}$$

Additionally, the MVU estimator for the linear model is efficient in that it attains the CRLB. The reader may now verify the result of (3.29) by substituting $\mathbf{H}$ for the line fitting problem into (4.4). The only detail that requires closer scrutiny is the invertibility of $\mathbf{H}^T\mathbf{H}$. For the line fitting example a direct calculation will verify that the inverse exists (compute the determinant of the matrix given in (3.29)). Alternatively, this follows from the linear independence of the columns of $\mathbf{H}$ (see Problem 4.2). If the columns of $\mathbf{H}$ are not linearly independent, as for example,

$$\mathbf{H} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ \vdots & \vdots \\ 1 & 1 \end{bmatrix}$$

and $\mathbf{x} = [2\,2\,\ldots\,2]^T$ so that $\mathbf{x}$ lies in the range space of $\mathbf{H}$, then *even in the absence of noise the model parameters will not be identifiable.* For then

$$\mathbf{x} = \mathbf{H}\boldsymbol{\theta}$$

and for this choice of $\mathbf{H}$ we will have for $x[n]$

$$2 = A + B \qquad n = 0, 1, \ldots, N-1.$$

As illustrated in Figure 4.1 it is clear that an infinite number of choices can be made for A and B that will result in the same observations or given a noiseless $\mathbf{x}$, $\boldsymbol{\theta}$ is not unique. The situation can hardly hope to improve when the observations are corrupted by noise. Although rarely occurring in practice, this degeneracy sometimes *nearly occurs* when $\mathbf{H}^T\mathbf{H}$ is ill-conditioned (see Problem 4.3).

The previous discussion, although illustrated by the line fitting example, is completely general, as summarized by the following theorem.

Theorem 4.1 (Minimum Variance Unbiased Estimator for the Linear Model) *If the data observed can be modeled as*

$$\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w} \tag{4.8}$$

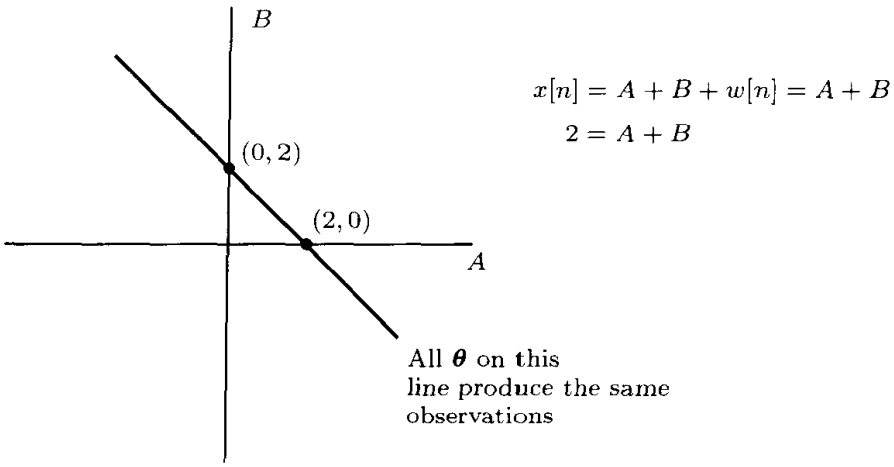


Figure 4.1 Nonidentifiability of linear model parameters

where $\mathbf{x}$ *is an* $N \times 1$ *vector of observations,* $\mathbf{H}$ *is a known* $N \times p$ *observation matrix (with* $N > p$*) and rank* p*,* $\boldsymbol{\theta}$ *is a* $p \times 1$ *vector of parameters to be estimated, and* $\mathbf{w}$ *is an* $N \times 1$ *noise vector with PDF* $\mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$*, then the MVU estimator is*

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x} \tag{4.9}$$

and the covariance matrix of $\hat{\boldsymbol{\theta}}$ *is*

$$\mathbf{C}_{\hat{\theta}} = \sigma^2(\mathbf{H}^T\mathbf{H})^{-1}. \tag{4.10}$$

For the linear model the MVU estimator is efficient in that it attains the CRLB.

That $\hat{\boldsymbol{\theta}}$ is unbiased easily follows by substituting (4.8) into (4.9). Also, the statistical performance of $\hat{\boldsymbol{\theta}}$ is *completely specified* (not just the mean and covariance) because $\hat{\boldsymbol{\theta}}$ is a *linear transformation* of a Gaussian vector $\mathbf{x}$ and hence

$$\hat{\boldsymbol{\theta}} \sim \mathcal{N}(\boldsymbol{\theta}, \sigma^2(\mathbf{H}^T\mathbf{H})^{-1}). \tag{4.11}$$

The Gaussian nature of the MVU estimator for the linear model allows us to determine the exact statistical performance if desired (see Problem 4.4). In the next section we present some examples illustrating the use of the linear model.

4.4 Linear Model Examples

We have already seen how the problem of line fitting is easily handled once we recognize it as a linear model. A simple extension is to the problem of fitting a curve to experimental data.

Example 4.1 - Curve Fitting

In many experimental situations we seek to determine an empirical relationship between a pair of variables. For instance, in Figure 4.2 we present the results of a experiment

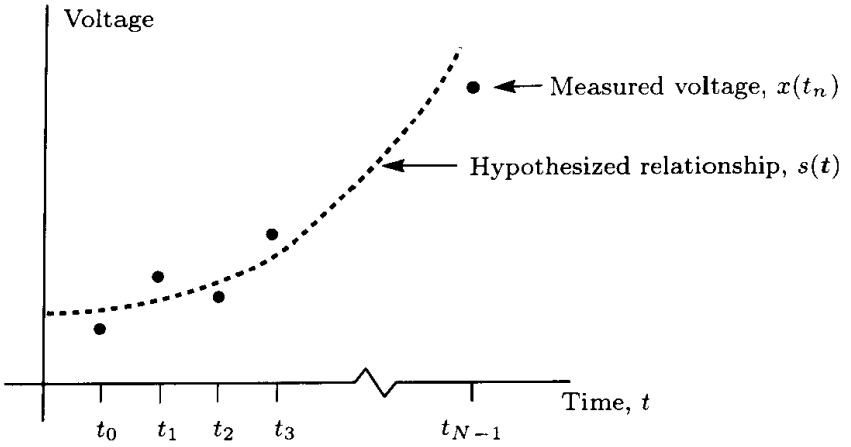


Figure 4.2 Experimental data

in which voltage measurements are taken at the time instants $t = t_0, t_1, \ldots, t_{N-1}$. By plotting the measurements it appears that the underlying voltage may be a quadratic function of time. That the points do not lie exactly on a curve is attributed to experimental error or noise. Hence, a reasonable model for the data is

$$x(t_n) = \theta_1 + \theta_2 t_n + \theta_3 t_n^2 + w(t_n) \qquad n = 0, 1, \ldots, N-1.$$

To avail ourselves of the utility of the linear model we assume that $w(t_n)$ are IID Gaussian random variables with zero mean and variance σ^2 or that they are WGN samples. Then, we have the usual linear model form

$$\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w}$$

where

$$\begin{aligned} \mathbf{x} &= [x(t_0)\, x(t_1) \ldots x(t_{N-1})]^T \\ \boldsymbol{\theta} &= [\theta_1\, \theta_2\, \theta_3]^T \end{aligned}$$

and

$$\mathbf{H} = \begin{bmatrix} 1 & t_0 & t_0^2 \\ 1 & t_1 & t_1^2 \\ \vdots & \vdots & \vdots \\ 1 & t_{N-1} & t_{N-1}^2 \end{bmatrix}.$$

In general, if we seek to fit a $(p-1)$st-order polynomial to experimental data we will have

$$x(t_n) = \theta_1 + \theta_2 t_n + \cdots + \theta_p t_n^{p-1} + w(t_n) \qquad n = 0, 1, \ldots, N-1.$$

The MVU estimator for $\boldsymbol{\theta} = [\theta_1\, \theta_2 \ldots \theta_p]^T$ follows from (4.9) as

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x}$$

where

$$\mathbf{x} = [x(t_0)\, x(t_1) \ldots x(t_{N-1})]^T$$

$$\mathbf{H} = \begin{bmatrix} 1 & t_0 & \ldots & t_0^{p-1} \\ 1 & t_1 & \ldots & t_1^{p-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & t_{N-1} & \ldots & t_{N-1}^{p-1} \end{bmatrix} \quad (N \times p).$$

The observation matrix for this example has the special form of a Vandermonde matrix. Note that the resultant curve fit is

$$\hat{s}(t) = \sum_{i=1}^{p} \hat{\theta}_i t^{i-1}$$

where $s(t)$ denotes the underlying curve or signal. ◇

Example 4.2 - Fourier Analysis

Many signals exhibit cyclical behavior. It is common practice to determine the presence of strong cyclical components by employing a Fourier analysis. Large Fourier coefficients are indicative of strong components. In this example we show that a Fourier analysis is really just an estimation of the linear model parameters. Consider a data model consisting of sinusoids in white Gaussian noise:

$$x[n] = \sum_{k=1}^{M} a_k \cos\left(\frac{2\pi kn}{N}\right) + \sum_{k=1}^{M} b_k \sin\left(\frac{2\pi kn}{N}\right) + w[n] \qquad n = 0, 1, \ldots, N-1 \quad (4.12)$$

where $w[n]$ is WGN. The frequencies are assumed to be harmonically related or multiples of the fundamental $f_1 = 1/N$ as $f_k = k/N$. The amplitudes a_k, b_k of the cosines and sines are to be estimated. To reformulate the problem in terms of the linear model we let

$$\boldsymbol{\theta} = [a_1\, a_2 \ldots a_M\, b_1\, b_2 \ldots b_M]^T$$

and

$$\mathbf{H} = \begin{bmatrix} 1 & \ldots & 1 & 0 & \ldots & 0 \\ \cos\left(\frac{2\pi}{N}\right) & \ldots & \cos\left(\frac{2\pi M}{N}\right) & \sin\left(\frac{2\pi}{N}\right) & \ldots & \sin\left(\frac{2\pi M}{N}\right) \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \cos\left[\frac{2\pi(N-1)}{N}\right] & \ldots & \cos\left[\frac{2\pi M(N-1)}{N}\right] & \sin\left[\frac{2\pi(N-1)}{N}\right] & \ldots & \sin\left[\frac{2\pi M(N-1)}{N}\right] \end{bmatrix}.$$

Note that $\mathbf{H}$ has dimension $N \times 2M$, where $p = 2M$. Hence, for $\mathbf{H}$ to satisfy $N > p$ we require $M < N/2$. In determining the MVU estimator we can simplify the computations by noting that the columns of $\mathbf{H}$ are orthogonal. Let $\mathbf{H}$ be represented in column form as

$$\mathbf{H} = [\mathbf{h}_1 \, \mathbf{h}_2 \ldots \mathbf{h}_{2M}]$$

where $\mathbf{h}_i$ denotes the ith column of $\mathbf{H}$. Then, it follows that

$$\mathbf{h}_i^T \mathbf{h}_j = 0 \qquad \text{for } i \neq j.$$

This property is quite useful in that

$$\begin{aligned} \mathbf{H}^T\mathbf{H} &= \begin{bmatrix} \mathbf{h}_1^T \\ \vdots \\ \mathbf{h}_{2M}^T \end{bmatrix} \begin{bmatrix} \mathbf{h}_1 & \ldots & \mathbf{h}_{2M} \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{h}_1^T\mathbf{h}_1 & \mathbf{h}_1^T\mathbf{h}_2 & \ldots & \mathbf{h}_1^T\mathbf{h}_{2M} \\ \mathbf{h}_2^T\mathbf{h}_1 & \mathbf{h}_2^T\mathbf{h}_2 & \ldots & \mathbf{h}_2^T\mathbf{h}_{2M} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{h}_{2M}^T\mathbf{h}_1 & \mathbf{h}_{2M}^T\mathbf{h}_2 & \ldots & \mathbf{h}_{2M}^T\mathbf{h}_{2M} \end{bmatrix} \end{aligned}$$

becomes a diagonal matrix which is easily inverted. The orthogonality of the columns results from the discrete Fourier transform (DFT) relationships for $i, j = 1, 2, \ldots, M < N/2$:

$$\begin{aligned} \sum_{n=0}^{N-1} \cos\left(\frac{2\pi in}{N}\right) \cos\left(\frac{2\pi jn}{N}\right) &= \frac{N}{2}\delta_{ij} \\ \sum_{n=0}^{N-1} \sin\left(\frac{2\pi in}{N}\right) \sin\left(\frac{2\pi jn}{N}\right) &= \frac{N}{2}\delta_{ij} \\ \sum_{n=0}^{N-1} \cos\left(\frac{2\pi in}{N}\right) \sin\left(\frac{2\pi jn}{N}\right) &= 0 \quad \text{for all } i, j. \end{aligned} \tag{4.13}$$

An outline of the orthogonality proof is given in Problem 4.5. Using this property, we have

$$\mathbf{H}^T\mathbf{H} = \begin{bmatrix} \frac{N}{2} & 0 & \ldots & 0 \\ 0 & \frac{N}{2} & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & \frac{N}{2} \end{bmatrix} = \frac{N}{2}\mathbf{I}$$

so that the MVU estimator of the amplitudes is

$$\begin{aligned}\hat{\boldsymbol{\theta}} &= (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x} \\ &= \frac{2}{N}\mathbf{H}^T\mathbf{x} = \frac{2}{N}\begin{bmatrix} \mathbf{h}_1^T \\ \vdots \\ \mathbf{h}_{2M}^T \end{bmatrix}\mathbf{x} \\ &= \begin{bmatrix} \frac{2}{N}\mathbf{h}_1^T\mathbf{x} \\ \vdots \\ \frac{2}{N}\mathbf{h}_{2M}^T\mathbf{x} \end{bmatrix}\end{aligned}$$

or finally,

$$\begin{aligned}\hat{a}_k &= \frac{2}{N}\sum_{n=0}^{N-1} x[n]\cos\left(\frac{2\pi kn}{N}\right) \\ \hat{b}_k &= \frac{2}{N}\sum_{n=0}^{N-1} x[n]\sin\left(\frac{2\pi kn}{N}\right).\end{aligned} \tag{4.14}$$

These are recognized as the discrete Fourier transform coefficients. From the properties of the linear model we can immediately conclude that the means are

$$\begin{aligned}E(\hat{a}_k) &= a_k \\ E(\hat{b}_k) &= b_k\end{aligned} \tag{4.15}$$

and the covariance matrix is

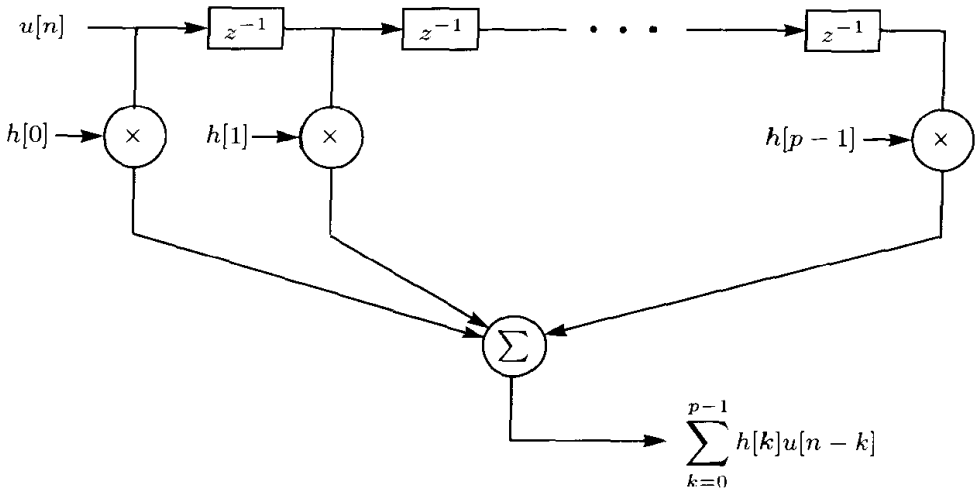
$$\begin{aligned}\mathbf{C}_{\hat{\theta}} &= \sigma^2(\mathbf{H}^T\mathbf{H})^{-1} \\ &= \sigma^2\left(\frac{N}{2}\mathbf{I}\right)^{-1} \\ &= \frac{2\sigma^2}{N}\mathbf{I}.\end{aligned} \tag{4.16}$$

Because $\hat{\boldsymbol{\theta}}$ is Gaussian and the covariance matrix is diagonal, the amplitude estimates are independent (see Problem 4.6 for an application to sinusoidal detection).

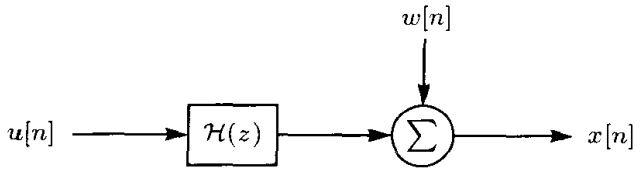
It is seen from this example that a key ingredient in simplifying the computation of the MVU estimator and its covariance matrix is the orthogonality of the columns of $\mathbf{H}$. Note that this property *does not hold* if the frequencies are arbitrarily chosen. ◇

Example 4.3 - System Identification

It is frequently of interest to be able to identify the model of a system from input/output data. A common model is the tapped delay line (TDL) or finite impulse response (FIR) filter shown in Figure 4.3a. The input $u[n]$ is known and is provided to "probe" the system. Ideally, at the output the sequence $\sum_{k=0}^{p-1} h[k]u[n-k]$ is observed from which



(a) Tapped delay line



$$\mathcal{H}(z) = \sum_{k=0}^{p-1} h[k]z^{-k}$$

(b) Model for noise-corrupted output data

Figure 4.3 System identification model

we would like to estimate the TDL weights $h[k]$, or equivalently, the impulse response of the FIR filter. In practice, however, the output is corrupted by noise, so that the model in Figure 4.3b is more appropriate. Assume that $u[n]$ is provided for $n = 0, 1, \ldots, N-1$ and that the output is observed over the same interval. We then have

$$x[n] = \sum_{k=0}^{p-1} h[k]u[n-k] + w[n] \qquad n = 0, 1, \ldots, N-1 \tag{4.17}$$

where it is assumed that $u[n] = 0$ for $n < 0$. In matrix form we have

$$\mathbf{x} = \underbrace{\begin{bmatrix} u[0] & 0 & \ldots & 0 \\ u[1] & u[0] & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ u[N-1] & u[N-2] & \ldots & u[N-p] \end{bmatrix}}_{\mathbf{H}} \underbrace{\begin{bmatrix} h[0] \\ h[1] \\ \vdots \\ h[p-1] \end{bmatrix}}_{\boldsymbol{\theta}} + \mathbf{w}. \tag{4.18}$$

Assuming $w[n]$ is WGN, (4.18) is in the form of the linear model, and so the MVU estimator of the impulse response is

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x}.$$

The covariance matrix is

$$\mathbf{C}_{\hat{\theta}} = \sigma^2(\mathbf{H}^T\mathbf{H})^{-1}.$$

A key question in system identification is how to choose the probing signal $u[n]$. It is now shown that the signal should be chosen to be pseudorandom noise (PRN) [MacWilliams and Sloane 1976]. Since the variance of $\hat{\theta}_i$ is

$$\text{var}(\hat{\theta}_i) = \mathbf{e}_i^T\mathbf{C}_{\hat{\theta}}\mathbf{e}_i$$

where $\mathbf{e}_i = [0\,0\ldots0\,1\,0\ldots0]^T$ with the 1 occupying the ith place, and $\mathbf{C}_{\hat{\theta}}^{-1}$ can be factored as $\mathbf{D}^T\mathbf{D}$ with $\mathbf{D}$ an invertible $p \times p$ matrix, we can use the Cauchy-Schwarz inequality as follows. Noting that

$$1 = (\mathbf{e}_i^T\mathbf{D}^T\mathbf{D}^{T^{-1}}\mathbf{e}_i)^2$$

we can let $\boldsymbol{\xi}_1 = \mathbf{D}\mathbf{e}_i$ and $\boldsymbol{\xi}_2 = \mathbf{D}^{T^{-1}}\mathbf{e}_i$ to yield the inequality

$$(\boldsymbol{\xi}_1^T\boldsymbol{\xi}_2)^2 \leq \boldsymbol{\xi}_1^T\boldsymbol{\xi}_1\boldsymbol{\xi}_2^T\boldsymbol{\xi}_2.$$

Because $\boldsymbol{\xi}_1^T\boldsymbol{\xi}_2 = 1$, we have

$$\begin{aligned} 1 &\leq (\mathbf{e}_i^T\mathbf{D}^T\mathbf{D}\mathbf{e}_i)(\mathbf{e}_i^T\mathbf{D}^{-1}\mathbf{D}^{T^{-1}}\mathbf{e}_i) \\ &= (\mathbf{e}_i^T\mathbf{C}_{\hat{\theta}}^{-1}\mathbf{e}_i)(\mathbf{e}_i^T\mathbf{C}_{\hat{\theta}}\mathbf{e}_i) \end{aligned}$$

or finally

$$\text{var}(\hat{\theta}_i) \geq \frac{1}{\mathbf{e}_i^T\mathbf{C}_{\hat{\theta}}^{-1}\mathbf{e}_i} = \frac{\sigma^2}{[\mathbf{H}^T\mathbf{H}]_{ii}}.$$

Equality holds or the minimum variance is attained if and only if $\boldsymbol{\xi}_1 = c\boldsymbol{\xi}_2$ for c a constant or

$$\mathbf{D}\mathbf{e}_i = c_i\mathbf{D}^{T^{-1}}\mathbf{e}_i$$

or, equivalently, the conditions for all the variances to be minimized are

$$\mathbf{D}^T\mathbf{D}\mathbf{e}_i = c_i\mathbf{e}_i \qquad i = 1, 2, \ldots, p.$$

Noting that

$$\mathbf{D}^T\mathbf{D} = \mathbf{C}_{\hat{\theta}}^{-1} = \frac{\mathbf{H}^T\mathbf{H}}{\sigma^2}$$

we have

$$\frac{\mathbf{H}^T\mathbf{H}}{\sigma^2}\mathbf{e}_i = c_i\mathbf{e}_i.$$

If we combine these equations in matrix form, then the conditions for achieving the minimum possible variances are

$$\mathbf{H}^T\mathbf{H} = \sigma^2 \begin{bmatrix} c_1 & 0 & \dots & 0 \\ 0 & c_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & c_p \end{bmatrix}.$$

It is now clear that to minimize the variance of the MVU estimator, $u[n]$ should be chosen to make $\mathbf{H}^T\mathbf{H}$ diagonal. Since $[\mathbf{H}]_{ij} = u[i-j]$

$$[\mathbf{H}^T\mathbf{H}]_{ij} = \sum_{n=1}^{N} u[n-i]u[n-j] \qquad i = 1, 2, \dots, p; j = 1, 2, \dots, p \tag{4.19}$$

and for N large we have (see Problem 4.7)

$$[\mathbf{H}^T\mathbf{H}]_{ij} \approx \sum_{n=0}^{N-1-|i-j|} u[n]u[n+|i-j|] \tag{4.20}$$

which can be recognized as a correlation function of the deterministic sequence $u[n]$. Also, with this approximation $\mathbf{H}^T\mathbf{H}$ becomes a symmetric Toeplitz autocorrelation matrix

$$\mathbf{H}^T\mathbf{H} = N \begin{bmatrix} r_{uu}[0] & r_{uu}[1] & \dots & r_{uu}[p-1] \\ r_{uu}[1] & r_{uu}[0] & \dots & r_{uu}[p-2] \\ \vdots & \vdots & \ddots & \vdots \\ r_{uu}[p-1] & r_{uu}[p-2] & \dots & r_{uu}[0] \end{bmatrix}$$

where

$$r_{uu}[k] = \frac{1}{N} \sum_{n=0}^{N-1-k} u[n]u[n+k]$$

may be viewed as an autocorrelation function of $u[n]$. For $\mathbf{H}^T\mathbf{H}$ to be diagonal we require

$$r_{uu}[k] = 0 \qquad k \neq 0$$

which is approximately realized if we use a PRN sequence as our input signal. Finally, under these conditions $\mathbf{H}^T\mathbf{H} = Nr_{uu}[0]\mathbf{I}$, and hence

$$\text{var}(\hat{h}[i]) = \frac{1}{Nr_{uu}[0]/\sigma^2} \qquad i = 0, 1, \dots, p-1 \tag{4.21}$$

and the TDL weight estimators are independent.

As a final consequence of choosing a PRN sequence, we obtain as our MVU estimator

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x}$$

where $\mathbf{H}^T\mathbf{H} = Nr_{uu}[0]\mathbf{I}$. Hence, we have

$$\begin{aligned}\hat{h}[i] &= \frac{1}{Nr_{uu}[0]} \sum_{n=0}^{N-1} u[n-i]x[n] \\ &= \frac{\frac{1}{N}\sum_{n=0}^{N-1-i} u[n]x[n+i]}{r_{uu}[0]} \end{aligned} \tag{4.22}$$

since $u[n] = 0$ for $n < 0$. The numerator in (4.22) is just the crosscorrelation $r_{ux}[i]$ between the input and output sequences. Hence, if a PRN input is used to identify the system, then the approximate (for large N) MVU estimator is

$$\hat{h}[i] = \frac{r_{ux}[i]}{r_{uu}[0]} \qquad i = 0, 1, \ldots, p-1 \tag{4.23}$$

where

$$\begin{aligned} r_{ux}[i] &= \frac{1}{N}\sum_{n=0}^{N-1-i} u[n]x[n+i] \\ r_{uu}[0] &= \frac{1}{N}\sum_{n=0}^{N-1} u^2[n]. \end{aligned}$$

◇

See also Problem 4.8 for a spectral interpretation of the system identification problem.

4.5 Extension to the Linear Model

A more general form of the linear model allows for noise that is not white. The *general linear model* assumes that

$$\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{C})$$

where $\mathbf{C}$ is not necessarily a scaled identity matrix. To determine the MVU estimator, we can repeat the derivation in Section 4.3 (see Problem 4.9). Alternatively, we can use a *whitening* approach as follows. Since $\mathbf{C}$ is assumed to be positive definite, $\mathbf{C}^{-1}$ is positive definite and so can be factored as

$$\mathbf{C}^{-1} = \mathbf{D}^T\mathbf{D} \tag{4.24}$$

where $\mathbf{D}$ is an $N \times N$ invertible matrix. The matrix $\mathbf{D}$ acts as a whitening transformation when applied to $\mathbf{w}$ since

$$\begin{aligned} E\left[(\mathbf{Dw})(\mathbf{Dw})^T\right] &= \mathbf{DCD}^T \\ &= \mathbf{DD}^{-1}\mathbf{D}^{T^{-1}}\mathbf{D}^T = \mathbf{I}. \end{aligned}$$

As a consequence, if we transform our generalized model

$$\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w}$$

to

$$\begin{aligned} \mathbf{x}' &= \mathbf{D}\mathbf{x} \\ &= \mathbf{D}\mathbf{H}\boldsymbol{\theta} + \mathbf{D}\mathbf{w} \\ &= \mathbf{H}'\boldsymbol{\theta} + \mathbf{w}' \end{aligned}$$

the noise will be whitened since $\mathbf{w}' = \mathbf{D}\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and the usual linear model will result. The MVU estimator of $\boldsymbol{\theta}$ is, from (4.9),

$$\begin{aligned} \hat{\boldsymbol{\theta}} &= (\mathbf{H}'^T\mathbf{H}')^{-1}\mathbf{H}'^T\mathbf{x}' \\ &= (\mathbf{H}^T\mathbf{D}^T\mathbf{D}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{D}^T\mathbf{D}\mathbf{x} \end{aligned}$$

so that

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{C}^{-1}\mathbf{x}. \tag{4.25}$$

In a similar fashion we find that

$$\mathbf{C}_{\hat{\theta}} = (\mathbf{H}'^T\mathbf{H}')^{-1}$$

or finally

$$\mathbf{C}_{\hat{\theta}} = (\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1}. \tag{4.26}$$

Of course, if $\mathbf{C} = \sigma^2\mathbf{I}$, we have our previous results. The use of the general linear model is illustrated by an example.

Example 4.4 - DC Level in Colored Noise

We now extend Example 3.3 to the colored noise case. If $x[n] = A + w[n]$ for $n = 0, 1, \ldots, N-1$, where $w[n]$ is *colored* Gaussian noise with $N \times N$ covariance matrix $\mathbf{C}$, it immediately follows from (4.25) that with $\mathbf{H} = \mathbf{1} = [1\,1\ldots1]^T$, the MVU estimator of the DC level is

$$\begin{aligned} \hat{A} &= (\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{C}^{-1}\mathbf{x} \\ &= \frac{\mathbf{1}^T\mathbf{C}^{-1}\mathbf{x}}{\mathbf{1}^T\mathbf{C}^{-1}\mathbf{1}} \end{aligned}$$

and the variance is, from (4.26),

$$\begin{aligned} \mathrm{var}(\hat{A}) &= (\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1} \\ &= \frac{1}{\mathbf{1}^T\mathbf{C}^{-1}\mathbf{1}}. \end{aligned}$$

If $\mathbf{C} = \sigma^2\mathbf{I}$, we have as our MVU estimator the sample mean with a variance of σ^2/N. An interesting interpretation of the MVU estimator follows by considering the factorization of $\mathbf{C}^{-1}$ as $\mathbf{D}^T\mathbf{D}$. We noted previously that $\mathbf{D}$ is a whitening matrix. The MVU estimator is expressed as

$$\begin{aligned}\hat{A} &= \frac{\mathbf{1}^T\mathbf{D}^T\mathbf{D}\mathbf{x}}{\mathbf{1}^T\mathbf{D}^T\mathbf{D}\mathbf{1}} \\ &= \frac{(\mathbf{D}\mathbf{1})^T\mathbf{x}'}{\mathbf{1}^T\mathbf{D}^T\mathbf{D}\mathbf{1}} \\ &= \sum_{n=0}^{N-1} d_n x'[n] \end{aligned} \tag{4.27}$$

where $d_n = [\mathbf{D}\mathbf{1}]_n/\mathbf{1}^T\mathbf{D}^T\mathbf{D}\mathbf{1}$. According to (4.27), the data are first prewhitened to form $x'[n]$ and then "averaged" using prewhitened averaging weights d_n. The prewhitening has the effect of decorrelating and equalizing the variances of the noises at each observation time before averaging (see Problems 4.10 and 4.11).

◇

Another extension to the linear model allows for signal components that are known. Assume that $\mathbf{s}$ represents a known signal contained in the data. Then, a linear model that incorporates this signal is

$$\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{s} + \mathbf{w}.$$

To determine the MVU estimator let $\mathbf{x}' = \mathbf{x} - \mathbf{s}$, so that

$$\mathbf{x}' = \mathbf{H}\boldsymbol{\theta} + \mathbf{w}$$

which is now in the form of the linear model. The MVU estimator follows as

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T(\mathbf{x} - \mathbf{s}) \tag{4.28}$$

with covariance

$$\mathbf{C}_{\hat{\theta}} = \sigma^2(\mathbf{H}^T\mathbf{H})^{-1}. \tag{4.29}$$

Example 4.5 - DC Level and Exponential in White Noise

If $x[n] = A + r^n + w[n]$ for $n = 0, 1, \ldots, N-1$, where r is *known*, A is to be estimated, and $w[n]$ is WGN, the model is

$$\mathbf{x} = A\begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} + \mathbf{s} + \mathbf{w}$$

where $\mathbf{s} = [1\, r \ldots r^{N-1}]^T$. The MVU estimator, is from (4.28),

$$\hat{A} = \frac{1}{N}\sum_{n=0}^{N-1}(x[n] - r^n)$$

with variance, from (4.29), as

$$\text{var}(\hat{A}) = \frac{\sigma^2}{N}.$$

◇

It should be clear that the two extensions described can be combined to produce the general linear model summarized by the following theorem.

Theorem 4.2 (Minimum Variance Unbiased Estimator for General Linear Model) *If the data can be modeled as*

$$\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{s} + \mathbf{w} \tag{4.30}$$

where $\mathbf{x}$ *is an* $N \times 1$ *vector of observations,* $\mathbf{H}$ *is a known* $N \times p$ *observation matrix* $(N > p)$ *of rank* p, $\boldsymbol{\theta}$ *is a* $p \times 1$ *vector of parameters to be estimated,* $\mathbf{s}$ *is an* $N \times 1$ *vector of known signal samples, and* $\mathbf{w}$ *is an* $N \times 1$ *noise vector with PDF* $\mathcal{N}(\mathbf{0}, \mathbf{C})$, *then the MVU estimator is*

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{C}^{-1}(\mathbf{x} - \mathbf{s}) \tag{4.31}$$

and the covariance matrix is

$$\mathbf{C}_{\hat{\theta}} = (\mathbf{H}^T\mathbf{C}^{-1}\mathbf{H})^{-1}. \tag{4.32}$$

For the general linear model the MVU estimator is efficient in that it attains the CRLB.

This theorem is quite powerful in practice since many signal processing problems can be modeled by (4.30).

References

Graybill, F.A., *Theory and Application of the Linear Model*, Duxbury Press, North Scituate, Mass., 1976.

MacWilliams, F.J., N.J. Sloane, "Pseudo-Random Sequences and Arrays," *Proc. IEEE*, Vol. 64, pp. 1715–1729, Dec. 1976.

Problems

4.1 We wish to estimate the amplitudes of exponentials in noise. The observed data are

$$x[n] = \sum_{i=1}^{p} A_i r_i^n + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $w[n]$ is WGN with variance σ^2. Find the MVU estimator of the amplitudes and also their covariance. Evaluate your results for the case when $p = 2, r_1 = 1, r_2 = -1$, and N is even.

4.2 Prove that the inverse of $\mathbf{H}^T\mathbf{H}$ exists if and only if the columns of $\mathbf{H}$ are linearly independent. Hint: The problem is equivalent to proving that $\mathbf{H}^T\mathbf{H}$ is positive definite and hence invertible if and only if the columns are linearly independent.

4.3 Consider the observation matrix

$$\mathbf{H} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1+\epsilon \end{bmatrix}$$

where ϵ is small. Compute $(\mathbf{H}^T\mathbf{H})^{-1}$ and examine what happens as $\epsilon \to 0$. If $\mathbf{x} = [2\,2\,2]^T$, find the MVU estimator and describe what happens as $\epsilon \to 0$.

4.4 In the linear model it is desired to estimate the signal $\mathbf{s} = \mathbf{H}\boldsymbol{\theta}$. If an MVU estimator of $\boldsymbol{\theta}$ is found, then the signal may be estimated as $\hat{\mathbf{s}} = \mathbf{H}\hat{\boldsymbol{\theta}}$. What is the PDF of $\hat{\mathbf{s}}$? Apply your results to the linear model in Example 4.2.

4.5 Prove that for $k, l = 1, 2, \ldots, M < N/2$

$$\sum_{n=0}^{N-1} \cos\left(\frac{2\pi kn}{N}\right) \cos\left(\frac{2\pi ln}{N}\right) = \frac{N}{2}\delta_{kl}$$

by using the trigonometric identity

$$\cos\omega_1 \cos\omega_2 = \frac{1}{2}\cos(\omega_1 + \omega_2) + \frac{1}{2}\cos(\omega_1 - \omega_2)$$

and noting that

$$\sum_{n=0}^{N-1} \cos\alpha n = \text{Re}\left(\sum_{n=0}^{N-1} \exp(j\alpha n)\right).$$

4.6 Assume that in Example 4.2 we have a single sinusoidal component at $f_k = k/N$. The model as given by (4.12) is

$$x[n] = a_k \cos(2\pi f_k n) + b_k \sin(2\pi f_k n) + w[n] \qquad n = 0, 1, \ldots, N-1.$$

Using the identity $A\cos\omega + B\sin\omega = \sqrt{A^2 + B^2}\cos(\omega - \phi)$, where $\phi = \arctan(B/A)$, we can rewrite the model as

$$x[n] = \sqrt{a_k^2 + b_k^2}\cos(2\pi f_k n - \phi) + w[n].$$

An MVU estimator is used for a_k, b_k, so that the estimated power of the sinusoid is

$$\hat{P} = \frac{\hat{a}_k^2 + \hat{b}_k^2}{2}.$$

A measure of detectability is $E^2(\hat{P})/\text{var}(\hat{P})$. Compare the measure when a sinusoid is present to the case when only noise is present or $a_k = b_k = 0$. Could you use the estimated power to decide if a signal is present?

4.7 Verify (4.20) by letting the limits of the summation in (4.19) be $n = -\infty$ to $n = \infty$ and noting that $u[n] = 0$ for $n < 0$ and $n > N - 1$.

4.8 We wish to estimate the frequency response $H(f)$ of a causal linear system based on the observed input and output processes. The input process $u[n]$ is assumed to be a WSS random process, as is the output $x[n]$. Prove that the cross-power spectral density between the input and output is

$$P_{ux}(f) = H(f)P_{uu}(f)$$

where $P_{ux}(f) = \mathcal{F}\{r_{ux}[k]\}$ and $r_{ux}[k] = E(u[n]x[n+k])$. If the input process is white noise with PSD $P_{uu}(f) = \sigma^2$, propose an estimator for the frequency response. Could you estimate the frequency response if $P_{uu}(f)$ were zero for a band of frequencies? Can you relate your estimator to (4.23) for the case when the linear system is a TDL?

4.9 Derive (4.25) and (4.26) by repeating the derivation in Section 4.3.

4.10 If in Example 4.4 the noise samples are uncorrelated but of unequal variance or

$$\mathbf{C} = \text{diag}(\sigma_0^2, \sigma_1^2, \ldots, \sigma_{N-1}^2)$$

find d_n and interpret the results. What would happen to $\hat{A}$ if a single σ_n^2 were equal to zero?

4.11 Assume the data model described in Problem 3.9. Find the MVU estimator of A and its variance. Why is the estimator independent of ρ? What happens if $\rho \to \pm 1$?

4.12 In this problem we investigate the estimation of a linear function of $\boldsymbol{\theta}$. Letting this new parameter be $\boldsymbol{\alpha} = \mathbf{A}\boldsymbol{\theta}$, where $\mathbf{A}$ is a known $r \times p$ matrix with $r < p$ and rank r, show that the MVU estimator is

$$\hat{\boldsymbol{\alpha}} = \mathbf{A}\hat{\boldsymbol{\theta}}$$

where $\hat{\boldsymbol{\theta}}$ is the MVU estimator of $\boldsymbol{\theta}$. Also, find the covariance matrix. Hint: Replace $\mathbf{x}$ by

$$\mathbf{x}' = \mathbf{A}(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x}$$

where $\mathbf{x}'$ is $r \times 1$. This results in the *reduced linear model.* It can be shown that $\mathbf{x}'$ contains all the information about $\boldsymbol{\theta}$ so that we may base our estimator on this lower dimensionality data set [Graybill 1976].

4.13 In practice we sometimes encounter the "linear model" $\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w}$ but *with* $\mathbf{H}$ *composed of random variables.* Suppose we ignore this difference and use our usual estimator

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x},$$

where we assume that the particular realization of $\mathbf{H}$ is known to us. Show that if $\mathbf{H}$ and $\mathbf{w}$ are independent, the mean and covariance of $\hat{\boldsymbol{\theta}}$ are

$$\begin{aligned} E(\hat{\boldsymbol{\theta}}) &= \boldsymbol{\theta} \\ \mathbf{C}_{\hat{\theta}} &= \sigma^2 E_H\left[(\mathbf{H}^T\mathbf{H})^{-1}\right] \end{aligned}$$

where E_H denotes the expectation with respect to the PDF of $\mathbf{H}$. What happens if the independence assumption is not made?

4.14 Suppose we observe a signal, which is subject to fading, in noise. We model the fading process as resulting in a signal that is either "on" or "off". As a simple illustration, consider the DC level in WGN or $x[n] = A + w[n]$ for $n = 0, 1, \ldots, N-1$. When the signal fades, the data model becomes

$$x[n] = \begin{cases} A + w[n] & n = 0, 1, \ldots, M-1 \\ w[n] & n = M, M+1, \ldots, N-1 \end{cases}$$

where the probability of a fade is ϵ. Assuming we know when the signal has experienced a fade, use the results of Problem 4.13 to determine an estimator of A and also its variance. Compare your results to the case of no fading.